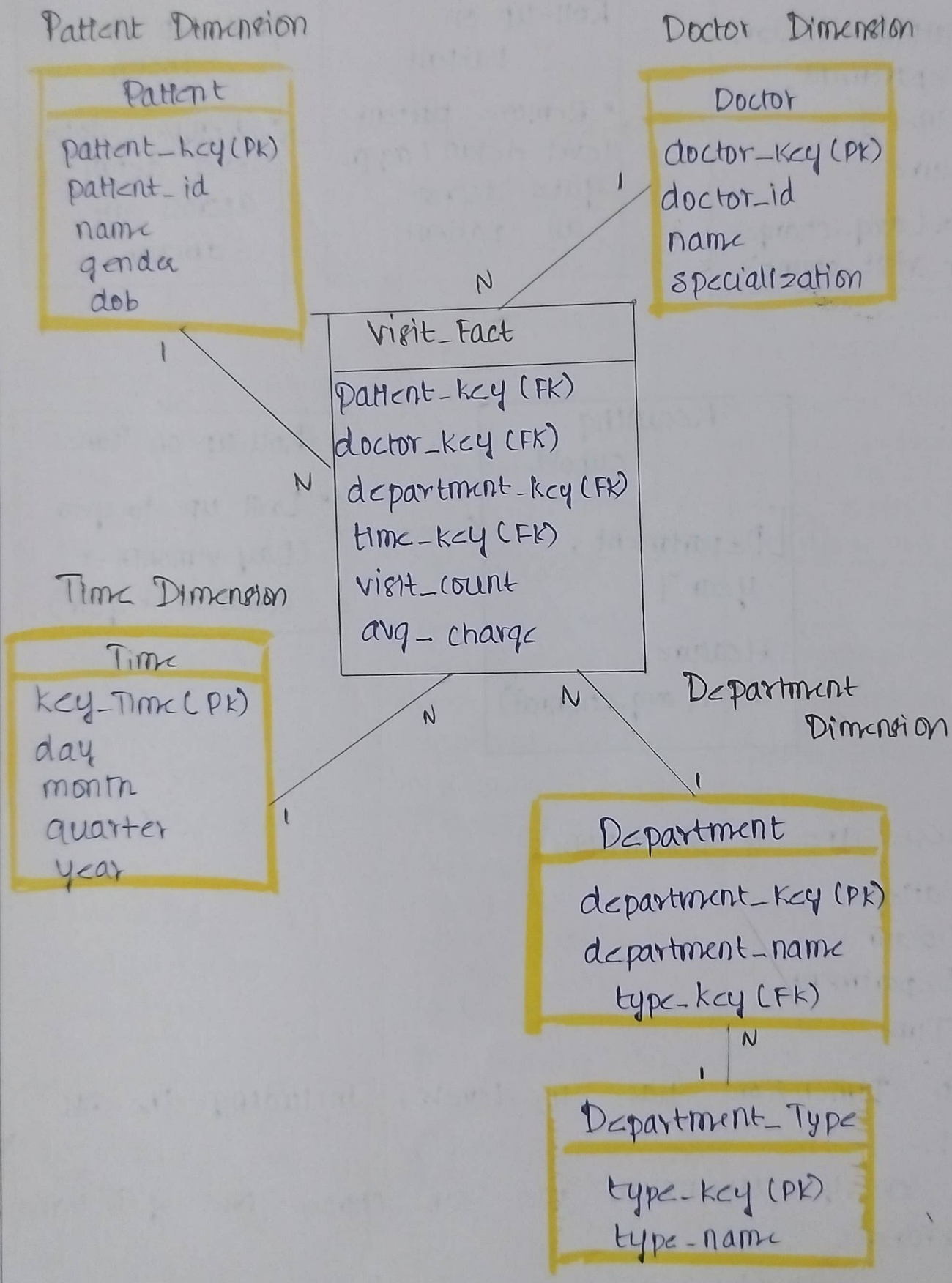
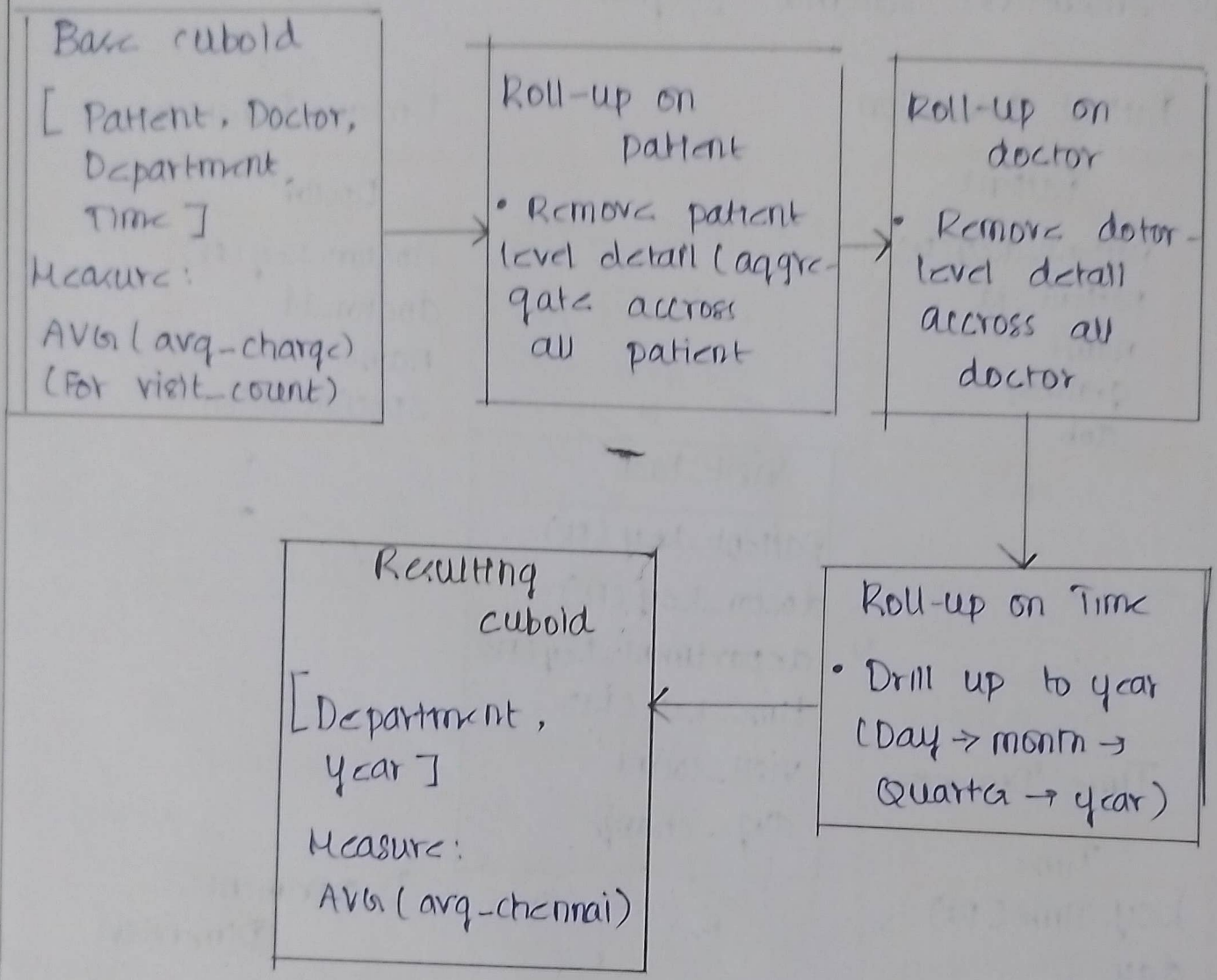
Name: Hemalakshmi .H
Reg.No: 14251049

2) Data Warehouse For city Hospital.

(a) SnowFlake schema Diagram



b) OLAP operations to get average consultation Fee department-wise for each year.



c) There are 4 Dimensions:

- Patient
- Doctor
- Department
- Time

Each Dimension has 4 levels, including the ALL level.

For every dimension, we can choose one of 4 levels.

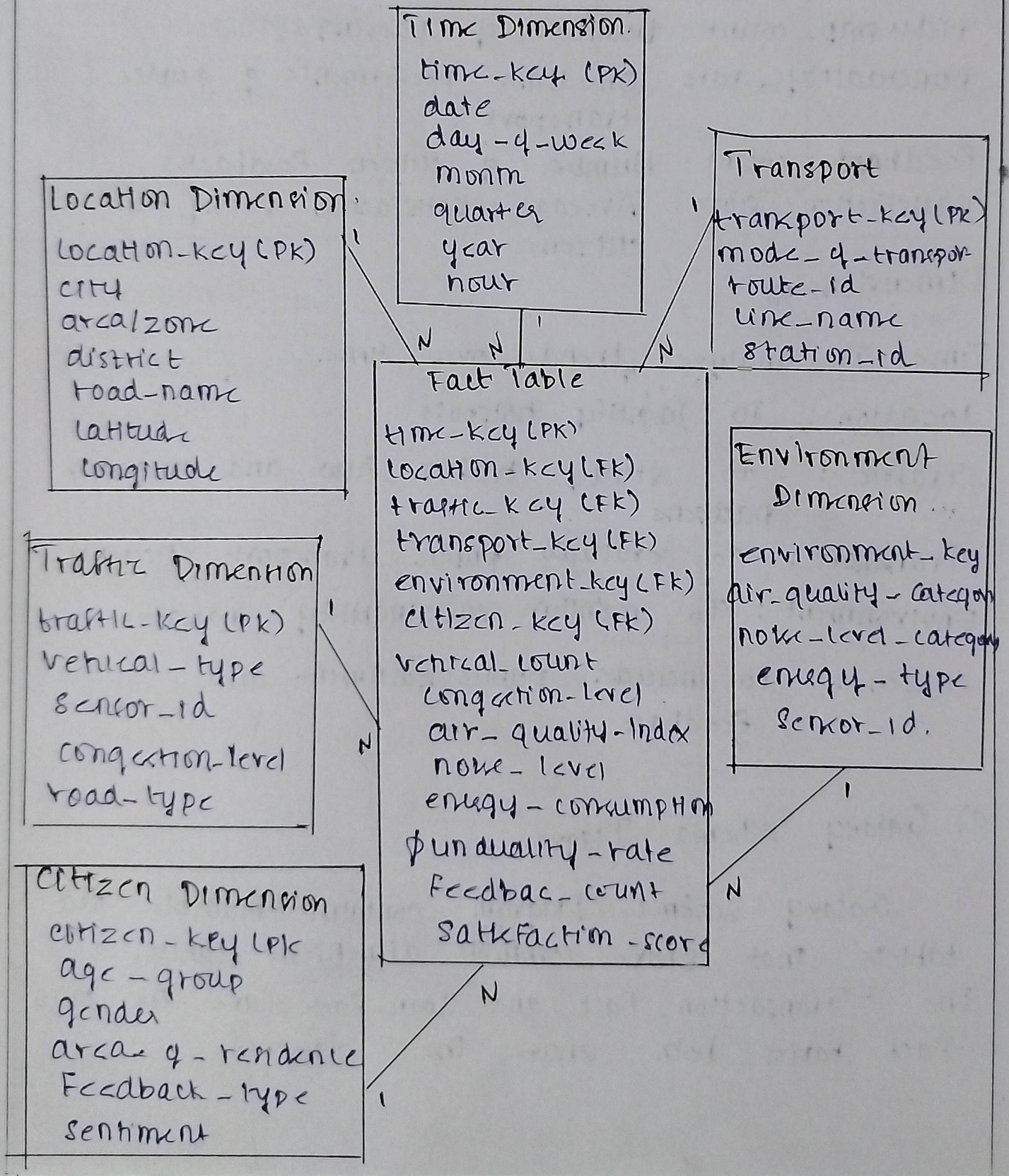
Therefore:

Number of cuboids = 4^4

= 256

- Since the cube has 4 Dimensions and each dimensions and has 4 levels including all level, the total number of cuboids is $4^4 = 256$ including the base cuboid and apex cuboid.

2. a) star schema For smart city Data Warehouse



b. Measures (Facts)

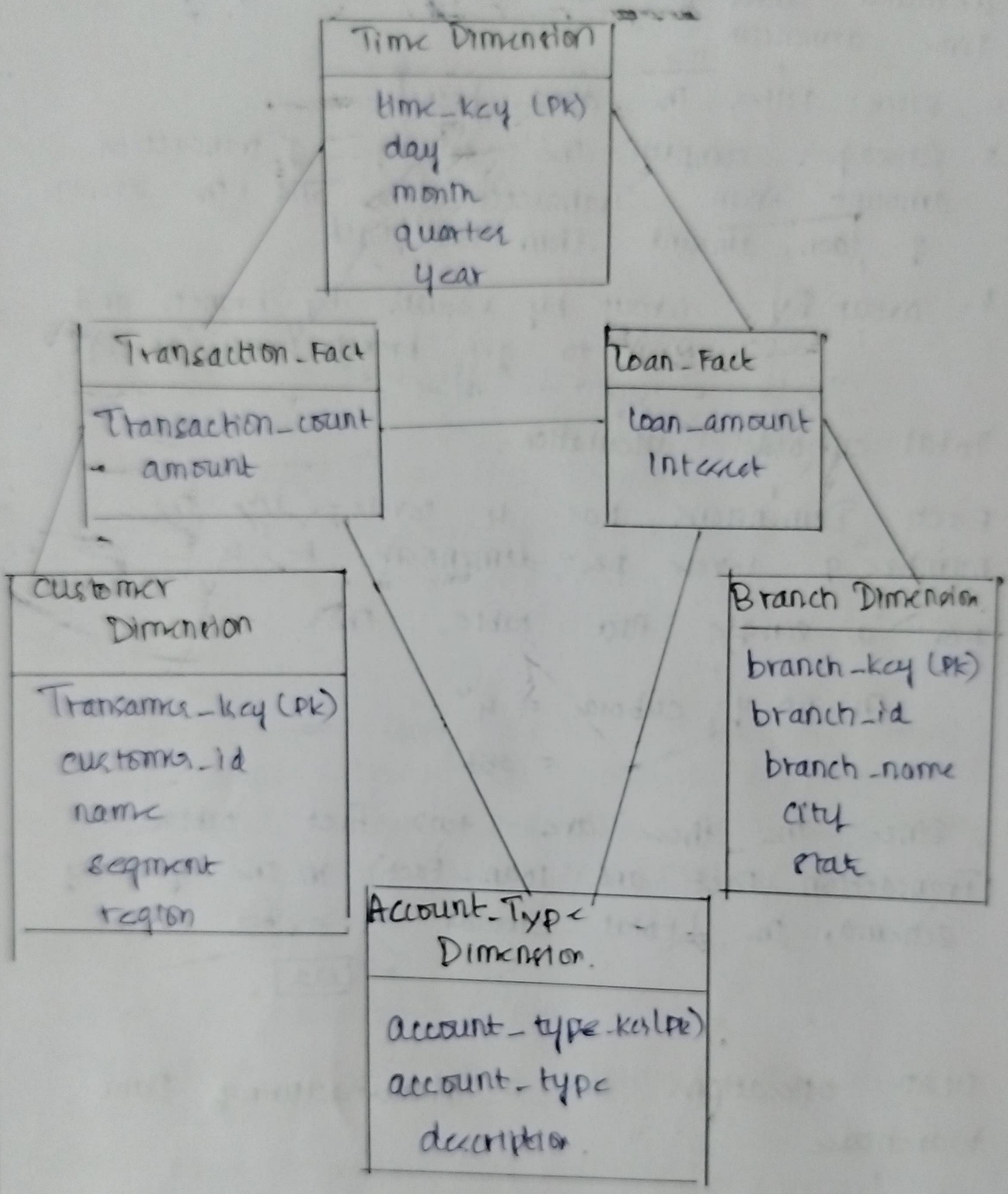
- Vehical_count : total number of vehicles
- congestion_level : level of traffic congestion
- air_quality_index : Air quality measurement
- noise_level : Noise level
- energy_consumption : Energy usage
- ridership_count : Number of passengers
- punctuality_rate : On-time performance of public transport
- Feedback_count : Number of citizen feedbacks
- satisfaction_score : Average satisfaction rating from citizens.

Dimensions:

- Time: To analyze trends over time.
- Location : To identify hotspots
- Traffic : To study traffic flow and congestion patterns
- Transport : To evaluate public transport efficiency
- Environment : To monitor sustainability indicators
- citizen : To include public sentiment and feedback.

5. a) Galaxy schema Diagram

A Galaxy schema Diagram contains multiple Fact tables that share common dimension tables. The Transaction_Fact and Loan_Fact are the two Fact tables, both share the dimensions



b OLAP operations

To obtain branch-wise average transaction amount and loan amount yearly, the following OLAP operations are used.

1. ROLL-UP (Drill-up): Aggregate data from the

detailed level to the highest level of year in the Time dimension.

2. Slice: Filters the data branch
3. Average: compute the average of transaction amount from Transaction_Fact and the average of loan amount from loan_Fact.
4. Group By: Group by results by branch and year to get branch wise average.

c. Total cuboids calculation:

Each Dimensions has 4 levels, let the number of level per dimensions be 4.

For a single fact table, the

the no. of cuboids $= 4^4$

$= 256$

Since the three are two fact tables (Transaction_Fact and loan_Fact) in the galaxy schema, the total cuboids $= 2 \times 256$

$= \boxed{512}$.

4) OLAP operations in a Manufacturing Data Warehouse

A Manufacturing company's data warehouse contains four dimensions: product, Factory, Time and supplier, along with measures such as Defect Rate, Production Units, and supplier Reliability Percentage.

1. Roll-up:

The data is aggregated from the lower levels of the product dimensions to the higher level product category. The Time dimension is rolled up to the year level, and data from all factories is consolidated.

Steps:

- Select the current year from Time dimension
- Roll up product from Batch → type - category.
- combine data from all factories.
- calculate the average Defect Rate for each product energy. category.

2. Drill-down:

Drill down moves from a higher level of detail to a lower level. The analysis is restricted to a specific Factory, and the data is examined at the Department level and product Type level.

Steps:

- Select a particular factory location.
- Drill down Factory from location → Department → line Numbers.
- Drill down product from category - Type.
- Analyze the Defect Rate for each department and product type.

3. Slice:

A slice selects a particular value or condition from one dimension while analyzing the remaining

dimension.

Steps:

- Select Suppliers whose rating is 4 or higher
- Include Suppliers from all regions
- Consider the production unit measure

4. Dice:

Dice select a subset of data using multiple Dimensions and conditions.

- Select the required product category
- Select the required Factory locations.
- Analyze the Defect Rate measure
- Analyze the Suppliers Reliability Percentage measure.

5) OLAP Operations in Retail Data Warehouse.

A retail chain's data warehouse contains Four dimension: Product, Store, Time and customer, along with measures such as Inventory Turnover, Total Sales Revenue, and units sold.

1. Roll-up:

Roll-up summarize data from a lower level to a higher level in a dimension hierarchy.

Steps:

- Select the last quarter from the Time dimension.
- Roll-up Product from SKU -> subcategory -> category.
- Include data from all stores.
- Summarize the Inventory Turnover for each produ

2. Drill - Down:

Drill Down moves from summarized Data to more detailed level.

Steps:

- Select Electronics from the product category.
- Select the required region from the store Dimension.
- Drill down Product from category - sub category - sku
- Analyze the Total sales Revenue and units sold at each level

3. Slice:

Slice selects data based on a specific condition from a dimension.

- Select Male from the Gender Dimension.
- Select Bronz from the Membership Tier
- Include data from all stores
- Analyze the Total sales Revenue Measure.

The sales revenue is obtained from for male customers belonging to the Bronze membership tier across all store.

4. Dice:

Dice selects a subset of the data cube using multiple dimension and conditions.

Steps:

- Select the required slices from the store Dimension.

- Select the required product categories from the product Dimension.
- select the required time periods from the Time Dimension.
- Analyze Total sales Revenue and units sold.

A smaller multidimensional data cube is obtained containing sales revenue and units sold for the selected cities, Product categories and time periods.